

MMS SECURITY, POPIA COMPLIANCE & DATA GOVERNANCE FRAMEWORK

Information Security, Privacy & Compliance Standards

Version 1.0

PURPOSE

This document establishes the security, privacy, compliance, governance, and data management standards for the Mpumalanga Mining Solutions Digital Ecosystem.

The objective is to:

Protect Student Data

Protect Business Data

Comply With South African Law

Maintain Trust

Reduce Risk

Support Future Growth

Ensure Responsible Use Of Technology

COMPLIANCE STANDARDS

The MMS platform must comply with:

POPIA (Protection of Personal Information Act)

PAIA (Promotion of Access to Information Act)

Electronic Communications And Transactions Act

South African Consumer Protection Principles

DATA GOVERNANCE PRINCIPLES

All data must be:

Lawfully Collected

Accurately Maintained

Securely Stored

Properly Processed

Auditable

Accessible Only To Authorized Users

Deleted When No Longer Required

DATA OWNERSHIP

Student Data belongs to the student.

MMS acts as a responsible custodian of student information.

Students may:

View Their Information

Update Their Information

Request Corrections

Request Certain Data Exports

Submit Privacy Requests

TYPES OF DATA COLLECTED

Personal Information

Full Names

ID Numbers

Date Of Birth

Gender

Nationality

Phone Numbers

Email Addresses

Residential Addresses

Emergency Contacts

Educational Information

Applications

Courses

Enrollments

Learning Progress

Assessments

Certificates

Attendance Records

Financial Information

Payment References

Invoices

Receipts

Proof Of Payment

Payment Status

System Information

Login History

Device Information

Portal Activity

Support Requests

Audit Logs

SENSITIVE DATA

The following information must receive enhanced protection:

ID Documents

Passport Copies

Proof Of Residence

Student Records

Payment Documents

Certificates

Administrative Records

DATA COLLECTION RULES

Before collecting information:

Explain Purpose

Obtain Consent

Provide Privacy Information

Allow User Review

Allow User Corrections

POPIA CONSENT REQUIREMENTS

Students must consent to:

Application Processing

Student Administration

Training Delivery

Communication

Certification Administration

Support Services

Compliance Requirements

MARKETING CONSENT

Marketing communication must be optional.

Students must have the ability to:

Opt In

Opt Out

Manage Preferences

Withdraw Consent

DATA STORAGE

Platform:

Supabase PostgreSQL

Storage:

Supabase Storage

Requirements:

Encryption

Backups

Access Control

Monitoring

Audit Trails

ACCESS CONTROL

Principle:

Least Privilege Access

Users only access information required for their role.

STUDENT ACCESS

Students may access:

Their Own Records

Their Own Documents

Their Own Certificates

Their Own Progress

Their Own Payments

INSTRUCTOR ACCESS

Instructors may access:

Assigned Students

Assigned Courses

Learning Data

Attendance Data

Support Data

ADMINISTRATOR ACCESS

Administrators may access:

Applications

Payments

Certificates

Student Records

Reports

Operational Data

SUPER ADMINISTRATOR ACCESS

Full System Access

Security Management

Role Management

System Configuration

Audit Review

PASSWORD POLICY

Minimum Length: 12 Characters

Requirements:

Uppercase

Lowercase

Number

Special Character

AUTHENTICATION

Supabase Authentication

Requirements:

Email Verification

Secure Password Storage

Session Management

Password Reset Workflow

FUTURE SECURITY

Two-Factor Authentication

Authenticator Apps

Device Verification

Login Alerts

DOCUMENT SECURITY

Uploaded documents must:

Be Encrypted

Be Access Controlled

Be Logged

Be Virus Checked

Be Securely Stored

FILE TYPES

Allowed:

PDF

PNG

JPG

JPEG

DOCX

Restricted:

Executable Files

Scripts

Unknown Formats

CERTIFICATE SECURITY

Every certificate must contain:

Unique Certificate Number

Issue Date

Verification Code

QR Verification

Digital Validation

CERTIFICATE VERIFICATION PAGE

Public Verification Allowed

Visitors may verify:

Certificate Number

Student Name

Course

Issue Date

Status

No sensitive information displayed.

AUDIT LOGGING

All critical actions must be logged.

TRACKED EVENTS

Login

Logout

Application Submission

Application Approval

Payment Verification

Certificate Issuance

Profile Updates

Document Uploads

Administrative Actions

AI Actions

DATA RETENTION

Applications: Minimum 5 Years

Certificates: Permanent Retention

Financial Records: Minimum 5 Years

Audit Logs: Minimum 3 Years

Support Records: Minimum 2 Years

BACKUP POLICY

Database Backups

Daily

Weekly

Monthly

DISASTER RECOVERY

Requirements:

Data Recovery Procedures

Backup Verification

Incident Response Plan

Recovery Testing

EMAIL SECURITY

Provider:

Resend

Requirements:

Verified Domains

SPF Records

DKIM Records

DMARC Records

Encryption In Transit

AI SECURITY

The MMS Assistant must:

Respect Permissions

Protect Student Data

Avoid Unauthorized Access

Avoid Hallucinations

Avoid Fabricated Information

Follow Escalation Procedures

AI RESTRICTIONS

The AI may not:

View Unauthorized Records

Modify Financial Data

Issue Certificates

Approve Applications

Override Security Controls

Reveal Private Information

DATA EXPORTS

Students may request:

Personal Information

Enrollment Information

Certificates

Learning Records

Subject To Verification

DATA DELETION REQUESTS

Students may request deletion where legally permissible.

Exceptions:

Financial Records

Compliance Records

Certificates

Legal Obligations

INCIDENT RESPONSE PLAN

If a security incident occurs:

Identify Incident

Contain Incident

Assess Impact

Notify Stakeholders

Recover Systems

Document Findings

Implement Improvements

PRIVACY NOTICE

The platform must clearly explain:

What Data Is Collected

Why It Is Collected

How It Is Used

Who Can Access It

How Long It Is Stored

How To Exercise Privacy Rights

THIRD PARTY SERVICES

Approved Services

Vercel

Supabase

Resend

NVIDIA APIs

Google Analytics

Microsoft Clarity

All integrations must be reviewed before implementation.

COMPLIANCE REPORTING

Administrators must be able to generate:

Privacy Reports

Access Reports

Audit Reports

Certificate Reports

Application Reports

Security Reports

EMPLOYEE RESPONSIBILITIES

All staff must:

Protect Student Information

Use Strong Passwords

Follow Security Policies

Report Security Incidents

Maintain Confidentiality

Follow POPIA Requirements

PLATFORM SECURITY SCORE TARGETS

Authentication Security: 95+

Data Security: 95+

Application Security: 95+

Infrastructure Security: 95+

Compliance Readiness: 95+

FINAL OBJECTIVE

The MMS platform must maintain the highest standards of privacy, security, compliance, and responsible data management.

Students must trust that their information is protected.

Administrators must operate confidently within a secure environment.

Every security decision must reinforce the reputation of Mpumalanga Mining Solutions as a professional, trustworthy, and responsible institution.